

UNIVERSITY MEDICAL CENTER

Department of Radiology – Division of Cardiovascular Imaging
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Examination: CARDIAC MRI WITH AND WITHOUT CONTRAST
Patient Name: Hick134, Brady998
Date of Birth: November 27, 1914
Age/Sex: 81 Years / Male
Medical Record #: MRN-4789621
Accession #: CMR19960808-5534
Exam Date/Time: August 8, 1996 at 14:30
Ordering Physician: Robert K. Chandler, MD – Cardiology
Referring Service: Cardiovascular Medicine / Critical Care

CLINICAL INDICATION

81-year-old male status post cardiac arrest on August 9, 1996. Patient successfully resuscitated with ROSC following VF arrest. Complex medical history including coronary artery disease with prior myocardial infarction, severe multi-vessel disease status post catheter ablation and biventricular ICD placement, diabetes mellitus, chronic kidney disease, recent stroke with residual monoparesis, and metastatic prostate carcinoma. Cardiac MRI requested for comprehensive evaluation of left ventricular systolic function, myocardial viability, scar burden, and assessment of ischemic substrate in the setting of recent life-threatening arrhythmia.

TECHNIQUE

MRI examination of the heart was performed on a Siemens Magnetom Avanto 1.5 Tesla scanner. The patient was positioned supine with a dedicated 32-channel cardiac phased-array coil. ECG gating was employed throughout the examination. Retrospective cardiac gating was utilized for cine imaging.

Sequences acquired include:

- Localizer sequences in three orthogonal planes
- Steady-state free precession (SSFP) cine imaging in standard short-axis stack (base to apex), two-chamber, three-chamber, and four-chamber long-axis views
- T2-weighted short-tau inversion recovery (STIR) imaging for myocardial edema assessment
- Native T1 mapping using Modified Look-Locker Inversion recovery (MOLLI) technique
- T2 mapping sequences
- Phase-contrast velocity imaging across the aortic valve
- Late gadolinium enhancement (LGE) imaging performed 10-15 minutes following intravenous administration of 0.15 mmol/kg gadobutrol (Gadavist) using phase-sensitive inversion recovery sequences in matching short-axis and long-axis orientations

Total examination time: 58 minutes. The patient tolerated the procedure well despite mild dyspnea. Heart rate ranged from 76-88 bpm with occasional premature ventricular contractions noted on ECG monitoring.

FINDINGS

Left Ventricle

Size and Function: The left ventricle is severely dilated with increased end-diastolic volume (EDV: 287 mL, indexed 162 mL/m²). The end-systolic volume is markedly increased (ESV: 207 mL, indexed 117 mL/m²). Left ventricular ejection fraction is severely reduced at approximately 28% by volumetric analysis. Stroke volume is diminished at 80 mL.

Regional Wall Motion: Severe hypokinesis to akinesis is present throughout the inferior and inferoseptal walls extending from base to mid-ventricular levels. The inferolateral wall demonstrates severe hypokinesis. The anterior and anterolateral walls show mild-to-moderate hypokinesis. The apical segments are severely hypokinetic with apical remodeling noted. No discrete left ventricular aneurysm is identified, though there is regional dyskinesis of the inferior wall.

Wall Thickness: End-diastolic wall thickness is reduced in the inferior and inferoseptal segments (5-6 mm), consistent with chronic transmural infarction and thinning. The anterior wall measures 9-10 mm. The septum demonstrates variable thickness with preserved thickness anteriorly (11 mm) and marked thinning inferiorly.

Late Gadolinium Enhancement: Extensive transmural late gadolinium enhancement (75-100% transmural extent) is present in the inferior wall, inferoseptal wall, and inferior aspect of the anteroseptal wall, extending from the basal to mid-ventricular levels. The pattern is consistent with chronic myocardial infarction in the distribution of the right coronary artery and/or left circumflex territory. Additional subendocardial to mid-myocardial LGE (25-50% transmural) is noted in the inferolateral wall. Focal subendocardial enhancement is seen in the basal anterior wall, suggesting additional prior subendocardial infarction. Total LGE burden is estimated at approximately 28% of left ventricular myocardium, indicating extensive non-viable myocardium with limited potential for functional recovery.

T2-weighted and Mapping Sequences: No significant myocardial edema is identified on T2-weighted imaging to suggest acute myocardial injury. Native T1 values are elevated in regions corresponding to LGE (1180-1240 ms), consistent with chronic scar. T2 values are within normal limits globally, confirming absence of acute inflammatory or edematous changes.

Right Ventricle

The right ventricle demonstrates mildly reduced systolic function with an estimated ejection fraction of 42%. Right ventricular end-diastolic volume is normal (142 mL, indexed 80 mL/m²). There is mild hypokinesis of the RV free wall. No late gadolinium enhancement is identified within the right ventricular myocardium. The RV outflow tract is patent without obstruction.

Atria

Both atria are moderately dilated. Left atrial volume is increased (98 mL, indexed 55 mL/m²). Right atrial enlargement is also present. No atrial thrombus is identified on post-contrast imaging.

Valves

The mitral valve demonstrates mild to moderate regurgitation based on cine imaging, with an eccentric posteriorly-directed jet, likely functional in etiology secondary to left ventricular dilatation and remodeling. The aortic valve is

trileaflet with mild sclerosis but no significant stenosis or regurgitation. The tricuspid valve shows trace regurgitation. The pulmonic valve appears structurally normal.

Pericardium

The pericardium is normal in thickness without evidence of effusion, thickening, or enhancement. No pericardial constriction.

Thoracic Aorta

The visualized thoracic aorta demonstrates age-appropriate atherosclerotic changes with mild ectasia of the ascending aorta (maximum diameter 3.8 cm). No aneurysm or dissection is identified.

Incidental Findings

Biventricular ICD leads are present and create expected susceptibility artifact. The device pocket is located in the left pectoral region. Small bilateral pleural effusions are noted. Mild dependent atelectasis is present in both lower lobes. Visualized upper abdomen demonstrates no acute abnormality on scout images.

IMPRESSION

- 1. SEVERELY REDUCED LEFT VENTRICULAR SYSTOLIC FUNCTION** with ejection fraction of 28%, consistent with advanced ischemic cardiomyopathy.
- 2. EXTENSIVE CHRONIC MYOCARDIAL INFARCTION** involving the inferior, inferoseptal, and inferolateral walls with transmural late gadolinium enhancement pattern consistent with RCA/LCx territory infarction. Total scar burden approximately 28% of LV myocardium represents predominantly non-viable tissue.
- 3. NO ACUTE MYOCARDIAL INJURY DETECTED.** Absence of myocardial edema on T2-weighted sequences and normal T2 mapping values argue against acute ischemic event coincident with the recent cardiac arrest.
- 4. SEVERE LEFT VENTRICULAR DILATATION AND REMODELING** with regional wall thinning in infarcted territories.
- 5. FUNCTIONAL MITRAL REGURGITATION**, mild to moderate in severity, secondary to LV dilatation and remodeling.
- 6. BIATRIAL ENLARGEMENT** and mildly reduced right ventricular systolic function (RVEF ~42%).
- 7. BIVENTRICULAR ICD** in place with leads visualized.

CLINICAL CORRELATION: The cardiac MRI findings confirm severe ischemic cardiomyopathy with extensive transmural scar and severely reduced left ventricular systolic function. The absence of acute myocardial edema suggests the recent cardiac arrest was likely arrhythmic in nature (ventricular fibrillation) arising from the extensive myocardial scar substrate rather than acute coronary occlusion. The large scar burden and severely reduced LVEF place the patient at continued high risk for malignant ventricular arrhythmias. The biventricular ICD is appropriately in place. Myocardial viability for potential revascularization is limited given the extensive transmural LGE pattern. Prognosis is guarded given the severity of LV dysfunction and extensive non-viable myocardium.

Report Finalized: August 8, 1996 at 17:45

Electronically Signed By:

Margaret L. Thornton, MD, FACR

Board Certified, Diagnostic Radiology

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This report has been electronically signed and is considered final. For questions regarding this report, please contact the Department of Radiology at (413) 555-7200.

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